

YINCHUAN, China

WMO: 536140

Lat: 38.4714N

Lon: 106.2078E

Elev: 1112

StdP: 88.66

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.8	-13.9	-25.7	0.4	-7.9	-23.4	0.5	-7.4	9.1	0.2	6.9	-2.0	1.7	0	0.444

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	32.9	18.7	31.7	18.5	30.4	18.0	22.0	28.4	21.2	27.7	20.4	26.9	2.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.1	17.0	25.4	19.2	16.0	24.9	18.2	15.0	24.2	70.9	28.3	67.4	27.8	64.2	26.8	26.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
7.7	6.0	4.9		-18.3	35.7	2.3	1.3	-20.0	36.6	-21.4	37.3	-22.7	38.0	-24.4	38.9	
				-19.0	23.3	2.2	1.5	-20.5	24.4	-21.8	25.2	-23.0	26.1	-24.6	27.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.6	-6.3	-1.7	5.6	13.2	18.2	22.6	24.6	22.9	17.8	10.9	2.9	-4.4
	DBStd	11.25	3.63	4.50	4.84	4.40	3.42	2.66	2.44	2.66	3.16	3.84	4.13	3.62
	HDD10.0	1699	504	328	151	19	1	0	0	0	0	35	216	446
	HDD18.3	3373	763	562	395	161	44	3	0	2	46	229	464	704
	CDD10.0	1913	0	0	14	115	255	378	452	399	234	64	1	0
	CDD18.3	546	0	0	0	7	41	131	194	142	30	1	0	0
	CDH23.3	4976	0	0	3	101	451	1228	1814	1142	224	13	0	0
	CDH26.7	1727	0	0	0	23	114	438	732	375	44	1	0	0
Wind	WSAvg	2.2	1.9	2.2	2.6	2.7	2.6	2.4	2.2	2.1	1.9	1.9	2.0	1.9
Precipitation	PrecAvg	198	1	2	6	11	20	20	42	47	26	13	4	1
	PrecMax	355	8	12	28	48	77	89	149	130	99	52	25	7
	PrecMin	77	0	0	0	0	0	0	2	4	1	0	0	0
	PrecStd	65	2	3	7	13	19	19	33	28	20	10	7	2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.4	13.5	22.6	29.3	31.5	34.0	35.2	33.6	30.7	25.1	17.1	8.7
		MCWB	0.0	3.8	8.8	12.8	14.4	16.6	19.6	19.7	17.1	12.6	6.9	1.0
	2%	DB	4.2	10.1	18.9	26.1	29.5	32.3	33.3	31.7	27.7	22.4	14.6	5.3
		MCWB	-1.6	2.1	7.2	11.1	13.9	16.8	19.6	19.6	15.8	11.9	5.9	-0.6
	5%	DB	2.1	7.9	16.4	23.7	27.8	30.8	32.0	30.3	25.8	20.2	12.2	3.4
		MCWB	-3.0	1.1	5.8	10.1	13.7	16.6	19.6	19.2	15.3	10.7	4.9	-1.5
	10%	DB	0.2	6.0	14.0	21.4	25.9	29.3	30.7	28.8	24.1	18.1	9.8	1.7
		MCWB	-4.1	-0.1	4.7	9.2	12.9	16.5	19.3	18.8	15.3	9.8	3.8	-2.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.2	4.6	9.7	14.5	17.2	20.7	23.5	23.2	20.4	14.8	8.4	1.6
		MCDB	6.9	11.3	19.6	23.1	26.2	28.0	30.6	29.3	24.1	21.7	14.0	7.2
	2%	WB	-1.5	2.7	7.9	12.8	16.0	19.6	22.4	21.9	18.6	13.4	6.7	-0.3
		MCDB	3.8	9.2	17.1	22.0	24.9	27.1	29.5	27.6	23.7	19.6	12.7	4.4
	5%	WB	-2.7	1.4	6.5	11.6	15.0	18.8	21.5	21.1	17.4	12.1	5.6	-1.3
		MCDB	1.7	7.3	15.1	20.3	23.7	26.3	28.5	27.1	22.7	17.8	10.9	3.0
	10%	WB	-4.0	0.1	5.2	10.4	14.1	18.0	20.7	20.3	16.3	10.9	4.3	-2.3
		MCDB	0.0	5.5	12.8	19.7	22.8	25.7	27.6	26.4	21.7	16.2	9.1	1.4
Mean Daily Temperature Range	5% DB	MDBR	10.6	11.9	11.9	12.4	12.1	11.2	10.3	10.0	10.4	11.7	9.8	9.4
		MCDBR	13.1	14.3	15.3	15.9	15.3	14.0	12.7	12.3	13.5	14.4	13.4	11.6
		MCWBR	8.6	8.4	7.4	6.2	5.6	4.5	4.0	4.0	4.8	6.1	6.9	7.3
	5% WB	MCDBR	12.6	13.7	14.1	12.8	12.1	11.1	10.2	9.3	9.9	12.2	12.0	10.8
		MCWBR	8.4	8.2	7.1	5.6	5.1	4.4	4.3	4.0	4.4	5.7	6.5	6.9
Clear-Sky Solar Irradiance	taub	0.353	0.410	0.464	0.504	0.522	0.529	0.510	0.513	0.492	0.446	0.382	0.345	
	taud	2.081	1.946	1.847	1.794	1.819	1.866	1.991	1.959	1.940	1.969	2.061	2.115	
	Ebn at Noon	788	786	783	780	775	768	780	765	750	737	745	765	
	Edh at Noon	117	155	189	212	211	201	176	178	171	149	119	106	
All-Sky Solar Radiation	RadAvg	2.83	3.76	5.00	6.00	6.66	6.57	6.20	5.59	4.62	4.06	3.10	2.60	
	RadStd	0.26	0.23	0.29	0.36	0.33	0.46	0.31	0.31	0.38	0.26	0.25	0.19	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.51	N/A	-1.23	+0.61	-0.51	-1.02	N/A	-149	+122	+63
(47)	Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	-0.58	N/A	N/A	N/A	N/A

Nomenclature: See separate page